

TED (15)3121

Reg No.....

REVISION-15

Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN WOOD & PAPER TECHNOLOGY

MECHANICAL AND ELECTRICAL ENGINEERING

MODEL QUESTION PAPER

Maximum marks: 100

Time: 3 hours

Note :- (PART 1 Mechanical Engineering and PART-2 Electrical Engineering to be answered in separate answer books)

PART-1 MECHANICAL ENGINEERING

(Maximum marks: 50)

PART-A

(Answer all questions in one or two sentences. Each question carries 2 marks)

(2x4=8marks)

- I.1. Define density
2. State dryness fraction of steam
3. State Bernoulli's theorem
4. What is boiler mountings

PART-B

(Answer any 2 questions. Each question carries 6 marks)

- II.1 Explain the construction and working of Venturimeter (6x2=12marks)

2. Explain belt drives
3. List out the advantages and disadvantages of chain drive over belt drive
4. Describe the classification of boilers

PART-C

(Answer one full question from each module. Each full question carries 15 marks) (15x2=30marks)

MODULE III

III.a) A simple U-tube manometer containing mercury is connected to a pipe in which a fluid of specific gravity 0.8 and having vacuum pressure is flowing. The other end of the manometer is open to the atmosphere. Find the vacuum pressure in pipe, if the difference of mercury level in the two limbs is 40cm and the height of fluid in the left from the centre of pipe is 15cm below

6

b) Explain construction and working of centrifugal pump with figure 9

OR

IV.a) A belt 100mm wide and 10mm thick is transmitting power at 1000meters/minute. The net driving tension is 1.8times the tension on the slack side. If the safe permissible stress on the belt section is 1.6Mpa, calculate the maximum power that can be transmitted at this speed. Assume density of leather is 1000kg/M³. Calculate the absolute maximum power that can be transmitted by this belt and the speed at which this can be transmitted 6

b) With the help of line diagram explain the working of a single acting reciprocating pump 9

MODULE IV

V.a) Explain working of a La-Mont boiler with line diagram 6

b) Compute the heat to be supplied to produce 1kg of super-heated steam with a temperature 230°C and at a pressure of 8bar from water at 20° C. 9

OR

VI.a) Describe boiler mountings 9

b) Compute the heat to be supplied per Kg of dry saturated steam at a pressure of 10 bar from water at 30° C 6

PART-2 ELECTRICAL ENGINEERING

(Maximum marks: 50)

PART-A

(Answer all questions in one or two sentences. Each question carries 2 marks)

(2x4=8marks) I.1. Define resistance and state its unit

2. Define electric current and state its unit

3. Name any two types of electron emission

4. What is synchronous speed of an induction motor

PART-B

(Answer any 2 questions. Each question carries 6 marks) (2x6=12marks)

II.1. Distinguish between self and mutual inductance

2. Explain the generation of ac

3. With neat sketches explain the construction of shell and core type transformer

4. Explain the construction and working of single phase induction type energy meter

PART-C

(Answer one full question from each module. Each full question carries 15 marks) (15x2=30marks)

MODULE III

III a) state and explain the laws of resistance

6

b) Three capacitors of $10\mu\text{F}$, $20\mu\text{F}$ and $30\mu\text{F}$ are connected 1) in series 2) in parallel. Compute the effective capacitance in each case

9

OR

IV.a) write the relationship between line and phase values of voltages and currents in star and delta connection

6

b) A capacitance of $20\mu\text{F}$ and a resistance of 100Ω are connected in series across 120V, 60Hz mains. Determine 1) capacitive reactance 2) impedance 3) current 4) power factor and 5) power consumed 9

MODULE IV

Va) Draw the constructional details of dc motor and mark its parts 7

b) A 10 KVA single phase 1100/230v transformer has a full load copper loss of 1200w and iron loss of 800 watts. Determine its efficiency at 0.8 p.f lagging at 1) full load and b) half load 8

OR

Vla) Draw a neat diagram of star-delta starter and mark its parts 7

b) With neat diagram explain the working of attraction type and repulsion type moving iron instrument 8